

**B.Tech. Degree III Semester Regular/Supplementary Examination
February 2022**

**ME 19-205-0304 FLUID MECHANICS
(2019 Scheme)**

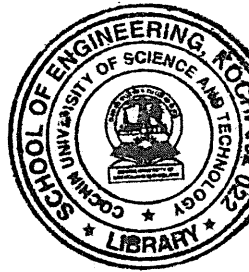
Time: 3 Hours

Maximum Marks: 60

PART A(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Explain Dynamic and Kinematic viscosity and the different units of these.
- (b) Explain the need of field approach in fluid flow analysis. Explain Lagrangian and Eulerian approach followed in fluid flow analysis.
- (c) Explain Venturi meter, Orifice meter and Flow nozzle meter.
- (d) What is meant by Hydraulic Grade Line (HGL) and Total Energy Line (TEL). Draw these lines when an incompressible fluid flows through a convergent divergent pipe.
- (e) Explain Source, Sink and Doublet.
- (f) Explain vorticity and circulation. Write the vorticity equation in terms of angular velocity.
- (g) Explain with a sketch the Boundary layer development in closed conduits.
- (h) What is meant by hydraulically rough and smooth pipes?

**PART B**

(4 × 12 = 48)

- II. (a) Two large plane surfaces are 30 mm apart. The space between the surface is filled with a lubricant of dynamic viscosity 8 poise. What force is required to drag a very thin plate of total surface area 0.8 square metre between the two large plane surfaces at a speed of 0.6m/s, if (7)
 - (i) the thin plate is in the middle of two plane surfaces
 - (ii) the thin plate is placed at one third the gap between the large plane surfaces.
- (b) A solid cylinder of diameter 4.0 m has a height of 4 m. Find the metacentric height of the cylinder if the specific gravity of the material of the cylinder = 0.6 and is floating in water with its axis vertical. State whether the equilibrium is stable or unstable. (5)

OR

- III. (a) Obtain the expression of excess pressure inside a water droplet, Soap bubble, and a liquid jet. (3)
- (b) The left leg of a U-tube mercury manometer is connected to a pipeline conveying water, the level of mercury in the leg being 0.8 m below the centre of pipe line and the right leg is open to atmosphere. The level of mercury in the right leg is 0.65 m above that in the left leg and the space above mercury in the right leg contains Benzene of sp gravity 0.88 to a height of 0.3 m. Find the pressure in the pipe. (5)
- (c) Find the magnitude and direction of the resultant force due to water acting on a roller gate of cylindrical form of 4 m diameter when the gate is placed on the dam in such a way that water is just going to spill. Take the length of the gate as 8 m. (4)



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- IV. (a) Derive Bernoulli's equation for a steady but rotational flow. (4)
 (b) Obtain an expression for absolute pressure at the vena contracta for an external mouth piece of cross sectional area 'a' and coefficient of contraction $C_c = 0.62$. (4)
 (c) Derive equation for the discharge through a Venturimeter. (4)

OR

- V. (a) State and prove Darcy-Weisbach equation. (4)
 (b) Derive the expression for loss of head due to sudden contraction. (3)
 (c) An oil of specific gravity 0.9 and viscosity 0.06 Poise is flowing through a pipe of diameter 200 mm at the rate of 60 lps. Find the head lost due to friction for a 500 m length of the pipe. Find the power required to maintain this flow. (5)

- VI. (a) Flow over a plane half body is studied by using a free stream flow at 6 m/s, superimposed on a source of strength 'q' at the origin. The body has a maximum width 3 m. calculate (6)
 (i) the coordinate of the stagnation point
 (ii) true width of the body at the origin
 (iii) velocity at a point $(0.5, \pi/2)$

- (b) A fluid flow field is given by $V = x^2 y i + y^2 z j - (2xyz + yz^2) k$. Prove that it is a case of possible steady incompressible fluid flow. Calculate the velocity and acceleration at the point (2,1,3). (6)

OR

- VII. (a) Prove that stream lines and equipotential lines cut orthogonally. Explain flownet. (4)
 (b) The velocity components in a two dimensional flow field for an incompressible field are expressed as $u = \frac{y^3}{3} + 2x - x^2 y, v = xy^2 - 2y - \frac{x^3}{3}$. (8)
 (i) Show that these functions represent a possible case of an irrotational flow
 (ii) Obtain expression for the stream function
 (iii) Obtain the expression for the velocity potential.

- VIII. (a) Find the displacement thickness, the momentum thickness and energy thickness for the velocity distribution in the boundary layer given by (6)

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2$$

- (b) Explain Boundary layer formation over a flat plate. Why is it necessary to control the growth of boundary layer on most of the bodies? What methods are used for such a control? (6)

OR

- IX. (a) Derive Von Karman momentum integral equation. (5)
 (b) A thin plate is moving in still atmospheric air at a velocity of 5 m/s. The length of the plate is 0.6 m and width 0.5 m. Calculate (7)
 (i) thickness of boundary layer at the end of the plate and
 (ii) drag force on one side of the plate. Take density of air as 1.24 kg/m^3

and kinematic viscosity 0.15 stokes. Take $\delta = \frac{4.91x}{\sqrt{\text{Re}_x}}$